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11 April 1975

Superseding
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MILITARY SPECIFICATION

SIGNAL KIT, PERSONNEL, DISTRESS, A/P25S-5A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements for one type of airborne, personnel, distress signal kit that is designated A/P25S-5A and that consists of a hand-fired launcher and of a bandoleer assembly containing seven, red signals (see 3.2.1 and 6.2). This specification also covers the requirements for spare bandoleer assemblies containing red, green, or white signals as specified (see 3.2.2 and 6.2).

2. APPLICABLE DOCUMENTS

* 2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

L-P-378	Plastic Sheet and Strip, Thin Gauge, Polyolefin
TT-I-548	Ink, Black, for Offset Duplicating Process
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-636	Boxes, Shipping, Fiberboard
PPP-F-320	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes

Military

MIL-P-116	Preservation-Packaging, Methods of
MIL-I-3606	Ink, Printing: Lithographic Offset, with Drier
MIL-C-5040	Cord, Nylon
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys

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STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-331	Fuze and Fuze Components, Environmental and Performance Tests for
MIL-STD-650	Explosive: Sampling, Inspection and Testing
MIL-STD-810	Environmental Test Methods
MIL-STD-882	Systems Safety Program for Systems and Associated Subsystems and Equipment: Requirements for
MIL-STD-889	Dissimilar Metals
MIL-STD-1167	Ammunition Data Cards
MIL-STD-1168	Lot Numbering of Ammunition

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

U. S. Department of Commerce

National Bureau of Standards Circular C-429	Photoelectric Tristimulus Colorimetry with Three Filters (R. R. Hunter, 1942)

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

Journal of the Optical Society of America

Dean B. Judd	The 1931 I. C. I. (International Commission on Illumination) Standard Observer and Coordinate System for Colorimetry (Volume 23, No. 10, 1933)

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George C. Sziklai A Tristimulus Photometer (Volume 41, No. 5, 1951)

(Application for copies should be addressed to the American Institute of Physics, 335 West 45th Street, New York, New York 10017.)

American Society for Testing and Materials (ASTM)

Symposium on Color - Specification and Use in Evaluating the Appearance of Materials (1941)

(Application for copies should be addressed to the University Microfilm Inc., 300 North Zeeb Road, Ann Arbor, Michigan 48103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries and are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Preproduction. This specification provides for preproduction testing (see 4.3 and 6.2).

3.2 Components

3.2.1 Signal kit. A signal kit shall consist of a launcher in a plastic bag, an instruction sheet, a plastic bag, and a bandoleer assembly. The bandoleer assembly of the signal kit shall consist of a plastic-molded bandoleer and seven red signals.

3.2.2 Spare bandoleer assembly. A spare bandoleer assembly shall consist of an instruction sheet, a plastic bag, a plastic-molded bandoleer, and seven signals. The seven signals in the spare bandoleer assembly shall be red, green, or white as specified (see 6.2).

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials

* 3.4.1 Nonmetals. Nonmetal material such as sealants and adhesives shall be a type that will not deteriorate with age or change properties to the extent that the signals fail to conform to the functional requirements specified under 3.7 at any time during the shelf or service life of the signal.

* 3.4.1.1 Age. Unless otherwise specifically approved by the procuring activity, all nonmetallic materials used in the firing mechanism, in the propulsion components, and in the signals shall have been manufactured not more than 18 months prior to the date of delivery to the Government.

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3.4.2 Metals. Metals shall be corrosion-resistant or shall be suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met in storage or normal use.

* 3.4.2.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in contact with each other. Dissimilar metals are defined in MIL-STD-889.

* 3.4.3 Compositions. All compositions (including but not limited to propellants, ignition compositions, delay compositions, transfer compositions, first fire compositions, and candle composition) shall be stable and shall be compatible with the other compositions and materials of construction.

3.5 Design and construction. The components of the signal kit and of the spare bandoleer assembly shall be designed and constructed to conform to the performance requirements specified herein.

* 3.5.1 Assembly of components

* 3.5.1.1 Assembly of signal kit. The components of the signal kit shall be assembled as follows: Install seven signals that are the same color (see 3.6.2) in the seven cavities of the bandoleer (see 3.6.3). Attach the launcher (see 3.6.1) to the bandoleer by utilizing two half hitches on the end of the lanyard. Place the bandoleer and the applicable instruction sheet (see 3.6.4) in the plastic bag (see 3.6.5). Force the excess air out of the plastic bag. Heat seal the plastic bag to provide a final closure.

* 3.5.1.2 Assembly of spare bandoleer assembly. The components of the spare bandoleer assembly shall be assembled as follows: Install seven signals that are the same color (see 3.6.2) in the seven cavities of the bandoleer (see 3.6.3). Place the bandoleer and the applicable instruction sheet (see 3.6.4) in the plastic bag (see 3.6.5). Force the excess air out of the plastic bag. Heat seal the plastic bag to provide a final closure.

* 3.5.3 Reliability. The signal and the launcher shall have a minimum functional reliability of 0.9547 at a lower confidence level of 90 percent.

* 3.5.4 Application of sealants and adhesives. Any sealants or adhesives used in the construction of the signal shall be applied in such a manner that the performance of the signal after prolonged storage shall not be degraded by aging or any other changes of the sealant or adhesive, e. g., a sealant used to seal the interface of the rocket motor and display case shall be applied in such a manner that excessive sealant will not enter the interior of the signal and cover or encapsulate the chemical components such as an igniter mix and in turn change the functional characteristics of the signal upon prolonged storage. Similarly, any sealant used to seal such components as the primer or interface between surfaces such as the nozzle plate and rocket motor housing, but not intended to seal the rocket motor nozzle ports closed, shall not be allowed to enter the nozzle ports.

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3.6 Details of components

* 3.6.1 Launcher. The launcher shall conform to figure 1 and shall hold and fire the signals specified in 3.6.2. The body of the launcher shall be aluminum. The distal end of the launcher shall have a collet-type, signal-gripping device and a firing mechanism that are made of corrosion-resistant steel. The firing mechanism shall consist of a free traveling firing pin with a smooth actuation knob and a spring. The proximal end of the launcher shall have a 30-inch-long lanyard attached to it. The lanyard shall be attached through a hole at the aft end, with the retaining knot on the inside, and shall be wrapped around the aft end of the launcher as shown in figure 1. The lanyard shall be olive drab or sage green and shall conform to type IA of MIL-C-5040. All threaded and press fitted parts of the launcher shall be held in place with a locking compound to preclude accidental disassembly or disassembly by hand. The launcher shall be placed, distal end first, in a bag.

* 3.6.1.1 Bag for launcher. The bag for the launcher shall be made of 0.002-inch-thick plastic that conforms to type I or II; class 1; grade A, B, or C; finish 1 or 2 of L-P-378. The bag for the launcher shall be open at one end and shall be 8-1/2 inches (plus or minus 1/8 inch) by 1-3/4 inches (plus or minus 1/8 inch).

3.6.2 Signals. Each signal (see 6.5) shall consist of a small solid-propellant-rocket motor actuated by a percussion primer, a delay element, and a pyrotechnic candle enclosed in a metal case and shall conform to figure 2. The colored surfaces specified in figure 2 shall be either painted with enamel or anodized in accordance with type I or II of MIL-A-8625 and dyed red, green, or white (see 3.11) to indicate that the color of the pyrotechnic candle in the signal is red, green, or white, respectively. The signal shall be spin stabilized by exhausting propellant gases through canted nozzles at the rear.

3.6.3 Bandoleer. The bandoleer shall be polyethylene molded to conform to figure 3. The color of the polyethylene shall be red, green, or white (see 3.11) to indicate that the color of the signals in the bandoleer is red, green, or white, respectively.

* 3.6.4 Instruction sheets. The instruction sheets shall be made of white Strathmore board (or equal), finish No. 7 on both sizes. (Strathmore board is manufactured by the Strathmore Paper Co., Industrial Speciality Papers Dept., 5 Front Street, West Springfield, Massachusetts.) The instruction sheets shall be 3-3/4 inches (plus or minus 1/8 inch) wide, 6-1/2 inches ((plus or minus 1/8 inch) long, and 0.008 inch (plus or minus 0.002 inch) thick. Unless otherwise specified herein, the instruction sheets shall be printed in approximately 1/8-inch bold type with black ink conforming to TT-I-548 or MIL-I-3606. The warning shall be in red ink in upper case letters in letter sizes and line spacing as indicated in figure 4. The information specified in 3.12.1 and the operating instructions shown on figure 4

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shall be included on both sides of the instruction sheet for the signal kit. The information specified in 3.12.2 and the operating instructions shown on figure 4 shall be included on both sides of the instruction sheet for the spare bandoleer assembly.

* 3.6.5 Plastic bag. Unless otherwise specified herein, the plastic bag shall conform to figure 5. As an option, the plastic bag may be a flat tube type of plastic bag with both ends heat sealed. The flat tube type of plastic bag shall conform to figure 5 except that the two sides shall not be heat sealed. The plastic bag shall be made of plastic conforming to type I or II; class 1; grade A, B, or C; finish 1 or 2 of L-P-378. The heat sealed seams specified on figure 5 shall be airtight and moisture proof.

3.7 Performance

3.7.1 Burn time, candlepower, and color

3.7.1.1 Burn time. A display with the following burn time (static candle) shall be produced by the pyrotechnic candle of the signal.

Display	Maximum Seconds	Minimum Seconds	Minimum Average Seconds
Red	12.5	8.0	9.0
Green	9.0	6.0	7.5
White	9.0	5.0	6.0

* 3.7.1.2 Candlepower. The pyrotechnic candle of the red signal shall produce a red display that has an average luminous intensity of 26,000 candle seconds and a minimum candlepower value of 2,500 candles for the first 8 seconds. The pyrotechnic candle of the green signal shall produce a green display that has a luminous intensity of a minimum total of 20,000 candlepower seconds. The pyrotechnic candle of the white signal shall produce a white display that has a luminous intensity of a minimum total of 16,000 candlepower seconds.

3.7.1.3 Color. The pyrotechnic candle of the red signal shall produce a red display that has a dominant wavelength of 620 millimicrons (plus or minus 20 millimicrons). The pyrotechnic candle of the green signal shall produce a green display that has a dominant wavelength of 540 millimicrons (plus or minus 20 millimicrons). The pyrotechnic candle of the white signal shall produce a white display that has a dominant wavelength of 575 millimicrons (plus or minus 10 millimicrons). The purity of the color of the red display produced by the pyrotechnic candle of the red signal shall be a minimum of 76 percent. The purity of the color of the green display produced by the pyrotechnic candle of the green signal shall be a minimum of 40 percent. The purity of the color of the white display produced by the pyrotechnic candle of the white signal shall be a maximum of 50 percent.

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* 3.7.2 Spin rate and velocity. Each signal shall have a minimum spin rate of 200 revolutions per second and a minimum velocity of 300 feet per second.

3.7.3 Function. The signals, when fired, shall function as follows:

* (a) The signal shall not be held captive in the launcher while the rocket motor burns. The signals shall not explode, deflagrate, or burn within 15 feet of the launcher. The components of the signal shall not separate or come apart within 15 feet of the launcher. Venting of the rocket motor which results in snuffing out of the propellant system after initial ignition shall not occur.

* (b) The candle shall produce a red, green, or white display in accordance with the color identification on the signal.

* (c) The burn time for each green display and for each white display shall be between 4 and 12 seconds. Except after the red signals have been subjected to temperature and humidity conditioning, the burn time of each red display shall be 7 to 15 seconds. The burn time of each red display after the red signals have been subjected to temperature and humidity conditioning shall be 5 to 15 seconds. Nonignition of the display shall be considered as zero burn time.

* (d) The apogee of the display, when launched vertically, shall be at least 700 feet but not more than 1,250 feet.

(e) The candle shall burn out prior to hitting the ground.

(f) The ignition of the candle shall not occur below 400 feet.

(g) The apogee of the display shall not deviate more than 30 degrees from vertical, using a vertically aligned launcher.

* 3.7.4 Safety functioning. When functioned as specified herein, the signals shall not explode, deflagrate, separate, burn, or become unsafe to handle. Signals that have not been damaged externally shall not malfunction in a hazardous manner (see 6.5) when functioned. Signals that have been damaged externally shall not be functioned but shall be safe to handle for disposal.

3.7.5 Penetration. The signals, when fired into a minimum of three each 1/4-inch-thick plywood panels, shall completely penetrate the panels. The signals, after penetrating the panels, shall produce a display of the appropriate color.

* 3.7.6 Jolt. The signals shall withstand jolt without burning, exploding, or becoming unsafe to handle. The signals shall not malfunction in a hazardous manner (see 6.5) when functioned.

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* 3.7.7 Jumble. The signals shall withstand jumble without burning, exploding, or becoming unsafe to handle. The signals shall not malfunction in a hazardous manner (see 6.5) when functioned.

* 3.7.8 Forty-foor drop. The signals shall withstand a 40-foot drop without burning, exploding, or becoming unsafe to handle. The signals shall not malfunction in a hazardous manner (see 6.5) when functioned.

* 3.7.9 Temperature and humidity. After the signals have been subjected to the temperature and the humidity conditioning specified herein, the signals shall function as specified in 3.7.12 and 3.7.13 as applicable.

3.7.10 Transportation vibration. The signals shall withstand transportation vibration without exploding or burning. After the signals have been subjected to transportation vibration, the signals shall function as specified in 3.7.3.

3.7.11 Altitude. After the signals have been placed in water in an altitude chamber and subjected to an altitude of 50,000 feet for 5 minutes, the signals shall function as specified in 3.7.3.

* 3.7.12 Low temperature. After the signals and the launcher have been stored at minus 65 degrees (plus or minus 5 degrees) Fahrenheit for a minimum of 5 hours, the launcher shall fire the signals. The signals shall function as specified in 3.7.3, except that the burn time of the display shall be 4 to 13 seconds and that the apogee of the signal shall be 400 feet or more.

* 3.7.13 High temperature. After the signals and the launcher have been stored at plus 160 degrees (plus or minus 5 degrees) Fahrenheit for a minimum of 5 hours, the launcher shall fire the signals. The signals shall function as specified in 3.7.3, except that the burn time of the display shall be 4 to 13 seconds and that the maximum height of the signal shall be not more than 1,600 feet.

3.7.14 Functioning after immersion in water. After the signals have been immersed in water to a depth of 11 to 13 inches of water for a minimum 3-hour period at a temperature of plus 70 degrees (plus or minus 10 degrees) Fahrenheit, the signals shall function as specified in 3.7.3.

* 3.7.15 Firing pin force (launcher). The firing pin force of the launcher shall be sufficient so that the launcher will initiate the primer. The firing pin force shall be a minimum of 6 pounds.

* 3.7.16 Ballistic functioning (launcher). The launcher shall fire a minimum of 14 signals without requiring maintenance of the launcher (no cleaning). The use of only one hand shall be required for the launcher to fire the signals.

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3.7.17 Vibration (launcher). After the launcher has been subjected to vibration at plus 86 degrees (plus or minus 18 degrees) Fahrenheit, the launcher shall fire a minimum of 14 signals without requiring maintenance of the launcher.

* 3.7.18 Salt spray (launcher). After the launcher has been exposed to salt spray for a 50-hour period, the launcher shall fire a minimum of 14 signals without requiring maintenance of the launcher.

* 3.7.19 Extreme high temperature storage. The signals shall withstand storage at a temperature of plus 200 degrees (plus or minus 5 degrees) Fahrenheit for a 14-day period without burning, exploding, or becoming unsafe to handle. The signals shall not malfunction in a hazardous manner (see 6.5) when functioned at ambient temperature.

* 3.7.20 High temperature storage. After the signals have been stored at plus 160 degrees (plus or minus 4 degrees) Fahrenheit for 28 days, the signals shall function as specified in 3.7.3 except that the burn time of the display shall be 4 to 12 seconds and that the maximum height of the display when launched shall be not more than 1,600 feet.

* 3.7.21 Low temperature storage. After the signals have been stored at minus 65 degrees (plus or minus 4 degrees) Fahrenheit for 28 days, the signals shall function as specified in 3.7.3 except that the burn time of the display shall be 4 to 12 seconds and that the maximum height of the display when launched shall be at least 400 feet.

3.8 System safety. A preliminary hazard analysis shall be conducted in accordance with MIL-STD-882 to determine that the storage and the operation of the signals conform to safety standards.

3.9 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.10 Weight. The weight of the launcher, including the lanyard, shall be 74 grams (plus or minus 10 grams). The weight of one red signal shall be 17.3 grams (plus or minus 3 grams). The weight of one green signal shall be 17.7 grams (plus or minus 3 grams). The weight of one white signal shall be 18.5 grams (plus or minus 3 grams).

3.11 Color. The color of the launcher, except for the collet-type, signal-gripping device; the firing pin; the spring; the actuation knob; and the lanyard, shall be black. All aluminum parts of the launcher shall be anodized and dyed a lusterless black color in accordance with type I or II of MIL-A-8625. The red, green, or white color used for the signals and the bandoleer (see 3.6.2 and 3.6.3) to indicate the color of the pyrotechnic

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candle shall conform approximately to 31136 or 31302 of FED-STD-595, 24260 of FED-STD-595, or 37875 of FED-STD-595, respectively.

3.12 Identification of product

* 3.12.1 Marking for signal kit. In addition to the operating instructions shown on figure 4, the following information shall be included on the instruction sheet for the signal kit.

Signal Kit, Personnel, Distress, A/P25S-5A
National Stock Number (NSN)
Lot No. of signal kits in accordance with MIL-STD-1168
Date of Manufacture (Month and Year)
Manufacturer's Identification and Contract No.

* 3.12.2 Marking for spare bandoleer assembly. In addition to the operating instructions shown on figure 4, the following information shall be included on the instruction sheet for the spare bandoleer assembly.

Bandoleer Assembly - Signal Flare, (Add Red, Green, or White as appropriate) for use with Signal Kit, Personnel, Distress, A/P25S-5A
NSN (as appropriate for red, green, or white)
Lot No. of bandoleers in accordance with MIL-STD-1168
Date of Manufacture (Month and Year)
Manufacturer's Identification and Contract No.

3.12.3 Marking for signal. Each signal shall be marked, in permanent ink, with the lot number and the date of manufacture in accordance with MIL-STD-1168.

3.12.4 Marking for launcher. Each launcher shall be marked, in accordance with MIL-STD-130, with the manufacturer's identification, the part number, and the contract number by imprint stamping.

3.12.5 Marking for bandoleer. Each bandoleer shall be marked with the manufacturer's identification and the lot number of the bandoleers.

* 3.13 Service life. The service life of the signal from date of manufacture shall be not less than 114 months, which shall consist of a shelf life of not less than 60 months plus an installed life of not less than 54 months.

* 3.14 Workmanship. The signal kit and the spare bandoleer assembly, in all stages of production, shall be processed under inspection control that will assure uniformity of quality. All components shall be free of chips, dirt, grit, or any other foreign matter. The cleaning methods used in manufacture shall not injure any of the parts. The cleaning agents employed shall not contaminate any of the parts.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The examination and the tests required by this specification are classified as follows:

(a) Preproduction tests

(b) Quality conformance tests.

4.3 Preproduction tests. The preproduction tests shall consist of the tests listed in table I and performed as described under 4.5. The preproduction samples listed in table I shall be representative of new untested items to be delivered on the contract or order and shall be subjected to the preproduction tests.

* 4.3.1 Preproduction samples. In addition to the preproduction samples required for the tests specified in table I, which are to be conducted by the contractor (see 4.1), untested preproduction samples that have been manufactured using the proposed full production methods shall be furnished by the contractor. Unless otherwise specified (see 6.2), the untested preproduction samples shall consist of two signal kits, two spare bandoleer assemblies with green signals, and two spare bandoleer assemblies with white signals. The signal kits and the spare bandoleer assemblies, including their components and all materials, shall be of single type, grade, class, size, and composition and shall comply with the applicable provisions of this specification. If any signal or launcher fails any preproduction test that has an acceptance criteria of "Ac 0 Re 1" (see table I) or if the apogee of any of the signals deviates more than 40 degrees from vertical (see 3.7.3(g)), all of the signals and launchers submitted for preproduction tests shall be rejected. Full particulars concerning the failures and action taken to correct the defect(s) shall be submitted to the procuring activity for concurrence and permission to repeat the preproduction tests. Any other failure(s) in the initial preproduction program which is outside the specified acceptance criteria shall require the submission of full particulars concerning the failure(s) and action taken to correct the defect(s) to the procuring activity for concurrence and permission to repeat the individual test(s) wherein the acceptance criteria was exceeded.

4.4 Quality conformance tests. The quality conformance tests shall consist of individual tests and sampling tests.

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* TABLE I. Preproduction Sample Tests

Test Paragraph	Samples to Be Tested	Requirement	Requirement Paragraph 1/	Acceptance Criteria 2/
4.5.2	15 red, 15 green, and 15 white signals	Burn time and candlepower	3.7.1.1 and 3.7.1.2	Ac 0 Re 1 3/
		Color	3.7.1.3	Ac 0 Re 1
4.5.3	5 red, 5 green, and 5 white signals	Spin rate and velocity	3.7.2	Ac 3 Re 4 4/
4.5.6	10 red, 10 green, and 10 white signals	Penetration	3.7.5	Ac 4 Re 5 5/
4.5.7	13 red, 13 green, and 13 white signals	Jolt	3.7.6	Ac 0 Re 1
4.5.5		Safety functioning	3.7.4	Ac 0 Re 1
4.5.8	13 red, 13 green, and 13 white signals	Jumble	3.7.7	Ac 0 Re 1
4.5.5		Safety functioning	3.7.4	Ac 0 Re 1
4.5.9	5 red, 5 green, and 5 white signals	Forty-foot drop	3.7.8	Ac 0 Re 1
4.5.5		Safety functioning	3.7.4	Ac 0 Re 1

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* TABLE I. Preproduction Sample Tests (Cont)

Test Paragraph	Samples to Be Tested	Requirement	Requirement Paragraph 1/	Acceptance Criteria 2/
4.5.10		Temperature and humidity	3.7.9	
	40 red signals		3.7.3(a) and (b)	Ac 0 Re 1
			3.7.3(c), (d), and (g) 6/	Ac 4 Re 5
			3.7.3(e) and (f)	Ac 6 Re 7
	40 green signals		3.7.3(a) and (b)	Ac 0 Re 1
			3.7.3(c), (d), and (g) 6/	Ac 4 Re 5
			3.7.3(e) and (f)	Ac 6 Re 7
	40 white signals		3.7.3(a) and (b)	Ac 0 Re 1
			3.7.3(c), (d), and (g) 6/	Ac 4 Re 5
			3.7.3(e) and (f)	Ac 6 Re 7

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TABLE I. Preproduction Sample Tests (Cont)

Test Paragraph	Samples to Be Tested	Requirement	Requirement Paragraph 1/	Acceptance Criteria 2/
4.5.11	7/	Transportation vibration	3.7.10	8/
4.5.12		Altitude	3.7.11	8/
	80 red signals 7/ 8/		3.7.3(a) and (b)	Ac 0 Re 1
			3.7.3(c), (d), and (g) 6/	Ac 4 Re 5
			3.7.3(e) and (f)	Ac 4 Re 5
	80 green signals 7/ 8/		3.7.3(a) and (b)	Ac 0 Re 1
			3.7.3(c), (d), and (g) 6/	Ac 4 Re 5
			3.7.3(e) and (f)	Ac 4 Re 5
	80 white signals 7/ 8/		3.7.3(a) and (b)	Ac 0 Re 1
			3.7.3(c), (d), and (g) 6/	Ac 4 Re 5
			3.7.3(e) and (f)	Ac 4 Re 5

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* TABLE I. Preproduction Sample Tests (Cont)

Test Paragraph	Samples to Be Tested	Requirement	Requirement Paragraph 1/	Acceptance Criteria 2/
4.5.13		Low temperature	3.7.12	
	40 red signals		3.7.3(a) and (b)	Ac 0 Re 1
			3.7.3(c), (d), and (g) 6/	Ac 4 Re 5
			3.7.3(e) and (f)	Ac 6 Re 7
	40 green signals		3.7.3(a) and (b)	Ac 0 Re 1
			3.7.3(c), (d), and (g) 6/	Ac 4 Re 5
			3.7.3(e) and (f)	Ac 6 Re 7
	40 white signals		3.7.3(a) and (b)	Ac 0 Re 1
			3.7.3(c), (d), and (g) 6/	Ac 4 Re 5
			3.7.3(e) and (f)	Ac 6 Re 7

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* TABLE I. Preproduction Sample Tests (Cont)

Test Paragraph	Samples to Be Tested	Requirement	Requirement Paragraph 1/	Acceptance Criteria 2/
4.5.14		High temperature	3.7.13	
	40 red signals		3.7.3(a) and (b)	Ac 0 Re 1
			3.7.3(c), (d), and (g) 6/	Ac 4 Re 5
			3.7.3(e) and (f)	Ac 4 Re 5
	40 green signals		3.7.3(a) and (b)	Ac 0 Re 1
			3.7.3(c), (d), and (g) 6/	Ac 4 Re 5
			3.7.3(e) and (f)	Ac 4 Re 5
	40 white signals		3.7.3(a) and (b)	Ac 0 Re 1
			3.7.3(c), (d), and (g) 6/	Ac 4 Re 5
			3.7.3(e) and (f)	Ac 4 Re 5

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TABLE I. Preproduction Sample Tests (Cont)

Test Paragraph	Samples to Be Tested	Requirement	Requirement Paragraph 1/	Acceptance Criteria 2/
4.5.15		Functioning after immersion in water	3.7.14	
	40 red signals		3.7.3(a) and (b)	Ac 0 Re 1
			3.7.3(c), (d), and (g) 6/	Ac 2 Re 3
			3.7.3(e) and (f)	Ac 2 Re 3
	40 green signals		3.7.3(a) and (b)	Ac 0 Re 1
			3.7.3(c), (d), and (g) 6/	Ac 2 Re 3
			3.7.3(e) and (f)	Ac 2 Re 3
	40 white signals		3.7.3(a) and (b)	Ac 0 Re 1
			3.7.3(c), (d), and (g) 6/	Ac 2 Re 3
			3.7.3(e) and (f)	Ac 2 Re 3

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* TABLE I. Preproduction Sample Tests (Cont)

Test Paragraph	Samples to Be Tested	Requirement	Requirement Paragraph 1/	Acceptance Criteria 2/
4.5.16	30 launchers	Firing pin force (launcher)	3.7.15	Ac 0 Re 1
4.5.17	10 launchers	Ballistic functioning (launcher)	3.7.16	Ac 0 Re 1
4.5.18	10 launchers	Vibration (launcher)	3.7.17	Ac 0 Re 1
4.5.19	10 launchers	Salt spray (launcher)	3.7.18	Ac 0 Re 1
4.5.20	10 red, 10 green, and 10 white signals	Extreme high temperature storage	3.7.19	Ac 0 Re 1

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* TABLE I. Preproduction Sample Tests (Cont)

Test Paragraph	Samples to Be Tested	Requirement	Requirement Paragraph 1/	Acceptance Criteria 2/
4.5.22	30 red, 30 green, and 30 white signals	High temperature storage	3.7.20	
			3.7.3(a) and (b)	Ac 0 Re 1
			3.7.3(c), (d), and (g) 6/	Ac 6 Re 7
			3.7.3(e) and (f)	Ac 6 Re 7
4.5.23	30 red, 30 green, and 30 white signals	Low temperature storage	3.7.21	
			3.7.3(a) and (b)	Ac 0 Re 1
			3.7.3(c), (d), and (g) 6/	Ac 6 Re 7
			3.7.3(e) and (f)	Ac 6 Re 7
4.5.24	10 red, 10 green, and 10 white signals	Weight check (signals and launchers)	3.10	Ac 1 Re 2
	10 launchers			Ac 1 Re 2

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TABLE I. Preproduction Sample Tests (Cont)

Test Paragraph	Samples to Be Tested	Requirement	Requirement Paragraph 1/	Acceptance Criteria 2/
4.5.26	MIL-STD-650	Compositions	3.4.3	MIL-STD-650

1/ The letters (a) through (g) opposite 3.7.3 pertain to the subparagraphs that contain the requirements to which the acceptance criteria are applicable.
2/ Any display that is not the color indicated by the color identification on the signal shall be cause for rejection of the entire preproduction sample submitted for test. The acceptance criteria shall apply to each color of signal, for example: If the column entitled "Samples to Be Tested" of this table specified 15 red, 15 green, and 15 white signals but only red signals are procured, 15 red signals shall be tested, and the acceptance criteria shall apply to the 15 tested red signals.
3/ The accept-reject criteria pertain to the minimum burn time. The results of 15 tests for each color shall be averaged to determine compliance with the minimum average burn time, the candlepower seconds, the dominant wavelength, and the color purity requirements specified in 3.7.1.1 and 3.7.1.2.
4/ The acceptance criteria pertain to the combination of spin rate and velocity determinations.
5/ Accept if four signals of each color fail to produce a display of the appropriate color after penetrating three boards. Reject if five signals of each color fail to produce a display of the appropriate color after penetrating three boards. If one or more signals fail to penetrate three boards, reject.
6/ Insofar as practical, the test for the requirements specified in 3.7.3(g) should be conducted when surface winds do not exceed 10 miles per hour. The apogee of the defective signals allowed by the acceptance criteria for this requirement shall not deviate more than 40 degrees from vertical. Reject if three samples deviate more than 30 degrees but less than 40 degrees from vertical. Reject if one sample deviates more than 40 degrees from vertical.
7/ Subject 40 of the signals to the transportation vibration test; subject 40 of the signals to the altitude test.
8/ The acceptance criteria for the transportation vibration and the altitude requirements are group data acceptance.

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* 4.4.1 Individual tests. Each launcher shall be subjected to the firing pin force (launcher) test described in 4.5.16.

4,4.2 Sampling tests

* 4.4.2.1 Signal kits. Samples of the production signal kits shall be selected in accordance with table I, special inspection level S-4 of MIL-STD-105 and shall be subjected to the examination of product described in 4.5.1. The unit of product (see MIL-STD-105) assembled into identifiable lots for the examination of product described in 4.5.1 shall be one complete signal kit (see 3.2.1), except that the plastic bag closure (see figure 5) shall not be heat sealed until after the examination of product. The acceptable quality level (AQL) shall be 0.25 percent for critical defects, 1.5 percent for major defects, and 4.0 percent for minor defects. Percent signifies percent defective.

* 4.4.2.2 Launchers and signals. Samples of the production launchers and signals shall be subjected to the tests specified in table II and described under 4.5. The acceptance criteria shall be as specified in table II. The unit of product (see MIL-STD-105) assembled into identifiable lots for all of the tests required by table II, except the ballistic functioning (launcher) described in 4.5.17 shall be one signal (see 3.6.2). The unit of product for the ballistic functioning (launcher) described in 4.5.17 shall be one launcher (see 3.6.1). The minimum lot size for signals (unit of product one signal as described in 3.6.2) shall be 3,201 signals except that the last lot may be less than 3,201 and the maximum lot size shall be 35,000 signals. The samples shall be selected in accordance with table II from the production lots except for the high temperature, the low temperture, and the temperature and humidity tests. Samples for the high temperature, the low temperature, and the temperature and humidity tests shall be selected from the initial and every fifth production lot only, i. e.: first lot, fifth lot, tenth lot, fifteenth lot, et cetera. If the initial or any fifth production lot fails to meet the acceptance criteria specified in table II, samples from the next production lot shall be subjected to the low temperature, the high temperature, and the temperature and humidity tests specified in table II and described under 4.5.

* 4.4.2.3 Resubmission of rejected lots. A rejected lot shall not be resubmitted for test and evaluation except with the approval of the contracting officer. The provisions listed under the paragraph entitled "Acceptance and Rejection" of MIL-STD-105 shall apply.

4.5 Test methods. Unless otherwise specified, the following test methods shall be conducted at ambient temperature, humidity, and barometric conditions.

4.5.1 Examination of product. The signal kit shall be examined, for the defects listed in table III, to determine conformance to the requirements of this specification.

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TABLE II. Production Sampling Tests

Inspection Level 1/	Test Paragraph	Requirement 2/	Requirement Paragraph 3/	AQL 4/	Classification of Defects 4/ 5/
S-3 of MIL-STD-105	4.5.2	Burn time and candlepower	3.7.1.1 and 3.7.1.2 6/	6.5%	Major
S-4 of MIL-STD-105	4.5.10	Temperature and humidity	3.7.9		
			3.7.3(a) and (b)	0.40%	Critical
			3.7.3(c), (d), and (g) 7/	4.0%	Major
			3.7.3(e) and (f)	6.5%	Minor
S-4 of MIL-STD-105	4.5.13	Low temperature	3.7.12		
			3.7.3(a) and (b)	0.40%	Critical
			3.7.3(c), (d), and (g) 7/	4.0%	Major
			3.7.3(e) and (f)	6.5%	Minor
S-4 of MIL-STD-105	4.5.14	High temperature	3.7.13		
			3.7.3(a) and (b)	0.40%	Critical
			3.7.3(c), (d), and (g) 7/	2.5%	Major
			3.7.3(e) and (f)	4.0%	Minor
II of MIL-STD-105	4.5.15	Functioning after immersion in water	3.7.14		
			3.7.3(a) and (b)	0.040%	Critical
			3.7.3(c), (d), and (g) 7/	1.5%	Major
			3.7.3(e) and (f)	2.5%	Minor

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TABLE II. Production Sampling Tests (Cont)

Inspection Level 1/	Test Paragraph	Requirement 2/	Requirement Paragraph 3/	AQL 4/	Classification of Defects 4/ 5/
S-4 of MIL-STD-105	4.5.17	Ballistic functioning (launcher)	3.7.16	1.5%	Major

1/ In the event different sample sizes are required, testing for all charactertics shall commence with the first sample unit tested and continue, in order, until the acceptance-rejection criterion for each AQL group has been satisfied. Upon completing the test requirements for the smaller sample size(s), subsequent testing shall be continued only for those characteristics requiring the larger sample size(s).
2/ The launchers used to perform the production sampling tests specified herein shall be production launchers.
3/ Except as indicated in 3.7.12 and 3.7.13, the letters (a) through (g) opposite 3.7.3 pertain to the subparagraphs that contain the requirements to which the acceptance criteria are applicable.
4/ The AQL stated shall be applied collectively for a given class (group) of defects, e. g., critical, major, or minor. Percent (%) signifies percent defective. Based upon allowable lot sizes and specified AQL for critical defects, if any critical defects occur, the lot shall be rejected.
5/ The defects to which this classification of defects apply shall be as listed in table IV.
6/ The requirement for minimum average burn time (see 3.7.1.1) shall not apply.
7/ Insofar as practicable, the tests for the requirements specified in 3.7.3(g) should be conducted when the surface winds do not exceed 10 miles per hour. The apogee of the defective signals allowed by the AQL specified for 3.7.3(g) shall not deviate more than 40 degrees from vertical. If one signal deviates more than 40 degrees from vertical, the lot shall be rejected.

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* TABLE III. Examination of Product

No.	Defects	Critical	Major	Minor
1	Canted nozzle at rear of signal omitted 1/	X		
2	Nozzle of signal incorrectly installed 1/	X		
3	Nozzle clogged with foreign matter 2/	X		
4	Launcher missing	X		
101	Signal dented 1/		X	
102	Loose actuation knob, collet, or handle of launcher 1/		X	
103	Firing pin interfering with collet 1/		X	
104	Color of signal assembly not as specified 3/		X	
105	Diameter of unpainted surface of signal not as specified 1/		X	
106	Primer of signal omitted or incorrectly located 1/		X	
107	Signal falls from collet when launcher is held with distal end pointed down		X	
108	Bandoleer containing less than 7 signals		X	
201	Instruction sheet missing			X
202	Knurling omitted from launcher or lanyard of launcher missing 1/			X
203	Burrs on launcher 1/			X
204	Plastic bag torn			X
205	Launcher not tied to bandoleer			X

1/ Inspection for this defect shall be performed prior to assembly.
2/ Nozzle holes should be visually inspected by using a 10 power miscroscope.
3/ This inspection shall pertain to the color of the signal only, for example, to determine that the signal is painted red when a red (display) signal is specified and not to determine the color shade of the signal.

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* TABLE IV. Classification of Defects

No.	Defects	Critical	Major	Minor
5	Launcher holds signal captive while rocket motor burns; signal explodes, deflagrates, or burns within 15 feet of launcher; or rocket motor vents and snuffs out propellant system	X		
6	Components of signal separate or come apart within 15 feet of launcher	X		
7	Color of candle display not in accordance with color identification on signal	X		
8	Apogee of signal deviates more than 40 degrees from vertical	X		
109	Maximum height of display not greater than 400 feet for signal that has been subjected to low temperature conditioning		X	
110	Maximum height of display less than 700 feet or more than 1,250 feet		X	
111	Maximum height of display exceed 1,600 feet after signal has been subjected to high temperature conditioning		X	
112	Launcher requires maintenance before launcher fires minimum of 14 signals		X	
113	Display fails to meet burn time and candlepower requirements of 3.7.1.1 and 3.7.1.2		X	
114	Apogee of display deviates more than 30 degrees but less than 40 degrees from vertical		X	
115	Burn time of display less than 4 seconds or more than 12 seconds for green or white signal 1/		X	

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* TABLE IV. Classification of Defects (Cont)

No.	Defects	Critical	Major	Minor
116	Burn time of display less than 7 seconds or more than 15 seconds for red signal (not conditioned) 1/		X	
117	Burn time of display less than 4 seconds or more than 13 seconds for signal that has not been subjected to low temperature conditioning 1/		X	
118	Burn time of display less than 5 seconds or more than 15 seconds for red signal that has been subjected to temperature and humidity conditioning 1/		X	
119	Burn time of display less than 4 seconds or more than 13 seconds for signal that has been subjected to high temperature conditioning 1/		X	
206	Display hitting ground while candle is still burning			X
207	Candle ignites below 400 feet			X

1/ Nonignition of the display shall be considered zero burn time.

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* 4.5.2 Burn time, candlepower, and color. The signal shall be disassembled. The pyrotechnic candle in its metal case shall be mounted in a test fixture in the fireplace of a photometer tunnel that has a nonreflecting surface. The signal shall be positioned vertically with the ignition end up and 2 inches below the photometric axis. A small amount of starter composition shall be placed on the ignition end of the signal. The starter composition shall be ignited with a resistance wire or electric match and a suitable power supply and shall be accomplished from outside the tunnel to assure safety of personnel. Light intensity and color shall be measured by the tristimulus photometer as described in the Journal of the Optical Society of America, Volume 41 (1951) pages 321 to 323, or equal. Calibration of the photometer shall be made using a projector lamp, calibrated for luminous intensity and color by the National Bureau of Standards, Washington, D. C. The luminous intensity, the dominant wavelength, and the percent of purity of the color produced by the burning signal assembly shall be measured and calculated, using the tristimulus photometer. The minimum candlepower value for the first 8 seconds shall be determined by the integration of the area under the candlepower time curve for the first 8 seconds then dividing by 8. The measurement and the calculation shall be made in accordance with the I. C. I. tristimulus color system as described in the Journal of the Optical Society of America, Volume 23, (1933), page 539; the ASTM Symposium on Color (1941); and the National Bureau of Standards Circular C-429. The burn time, the candlepower seconds, the dominant wavelength, and the percent purity shall be determined from the recording. Signal assemblies that fail to meet the requirements specified in 3.7.1.1 through 3.7.1.3 shall be considered to be defectives.

4.5.3 Spin rate and velocity. The tests to determine the spin rate and the velocity of the signals shall be conducted at ambient temperatures. The spin rate shall be determined between 12 and 18 feet from the launch site and shall be determined by mathematical computations or empirically. The velocity shall be determined between 12 and 18 feet from the launch site and shall be determined by utilizing an electronic counter-type chronograph measuring to at least the nearest microsecond. Either high velocity Lumiline screens or electrical contact screens that either open or close an electric circuit by passage of the signal through the detector shall be utilized. Contact screens may consist either of metallic foils separated by a thin insulating layer or of a circuit printed on paper with the circuit spacing such that the signal passing through the screen will "break" the circuit. When prior approval is granted by the procuring activity, optional methods of measuring velocity may be utilized.

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* 4.5.4 Function. Only launchers, as applicable, meeting the requirements of this specification shall be used to test the signals. (CAUTION: During handling or testing, the launcher with the assembled signal shall not be pointed toward personnel.) The launcher shall be hand initiated (with only one hand), signal end up in a true vertical position, and held a minimum of 12 inches above the head or remotely in a suitable test fixture. A minimum of ten signals shall be hand launched (with only one hand). A maximum of two attempts to initiate the signal shall be made, except that a maximum of four attempts shall be made with signals that have been subjected to cold temperature. Signal exploding, component separation, signal held captive in launcher while rocket motor burns, signal burning within 15 feet of the launcher, maximum height of signal, duration of the burn time, ignition height, vertical deflection, ignited signal hitting the ground, and any signal failing to produce the proper color display shall be measured and recorded. Preproduction signals and production signals that fail to meet the requirements of tables I and II, respectively, shall be considered to be defective signals.

4.5.5 Safety functioning. The signals that are subjected to the test to determine safety functioning shall not be dented or damaged in any manner that will affect functioning. The signals shall be fired remotely utilizing the launcher specified herein but modified for remote actuation. The height of the ejection of the signals and the exploding of any of the signals shall be recorded. Signals that fail to meet the requirements specified in 3.7.4 shall be considered to be defectives.

4.5.6 Penetration. The signals shall be fired into a fixture located 25 feet from the launch site. The fixture shall consist of three exterior, commercial grade plywood panels that are spaced 3-5/8 inches apart. Each of the three plywood panels shall be 4 feet long by 4 feet wide by 1/4 inch thick. After the signals penetrate the plywood, the signals shall produce a display of the appropriate color.

4.5.7 Jolt. The signals shall be subjected to the jolt test described in test 101 of MIL-STD-331 using a suitable jolt fixture. Signals that burn, explode, or become unsafe to handle shall be considered to be defectives. After the jolt test, the signals shall be subjected to the safety functioning test described in 4.5.5.

4.5.8 Jumble. The signals shall be subjected to the jumble test described in test 102 of MIL-STD-331. Signals that burn, explode, or become unsafe to handle shall be considered to be defectives. After the jumble test, the signals shall be subjected to the safety functioning test described in 4.5.5.

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4.5.9 Forty-foot drop. The signals shall be subjected to the 40-foot drop test described in test 103 of MIL-STD-331, except that the signals shall be dropped free fall with no guidance. Signals that burn, explode, or become unsafe to handle shall be considered to be defectives. After the 40-foot drop test, the signals shall be subjected to the safety functioning test described in 4.5.5.

* 4.5.10 Temperature and humidity. The signals shall be subjected to one 14-day temperature and humidity cycle as described in test 105 of MIL-STD-331. At the conclusion of the 14-day temperature and humidity cycle, 50 percent of the signals shall be conditioned at plus 160 degrees Fahrenheit for a minimum of 5 hours; the other 50 percent of the signals shall be conditioned at minus 65 degrees Fahrenheit for a minimum of 5 hours. The signals shall be functioned within 30 seconds after removal from the chamber. Signals that fail to meet the requirements specified in 3.7.13 after the conditioning at plus 160 degrees Fahrenheit shall be considered to be defective. Signals that fail to meet the requirements specified in 3.7.12 after the conditioning at minus 65 degrees Fahrenheit shall be considered to be defective.

4.5.11 Transportation vibration. The signals shall be subjected to the transportation vibration test at plus 86 degrees (plus or minus 18 degrees) Fahrenheit as described in test 104, procedure II of MIL-STD-331 using a suitable test fixture. After the completion of the transportation vibration test, the signal shall be immersed in 11 to 13 inches of water for 1 hour. The signals shall then be functioned as described in 4.5.4. Signals that fail to function as specified in 3.7.3 shall be considered to be defectives.

4.5.12 Altitude. The signals shall be placed in water in an altitude chamber and subjected to an altitude of 50,000 feet for 5 minutes. Signals that show bubbles shall be considered defectives. The signals shall then be functioned as described in 4.5.4. Signals that fail to function as specified in 3.7.3 shall be considered to be defectives.

4.5.13 Low temperature. The signals and the launcher shall be soaked a minimum of 5 hours at minus 65 degrees (plus or minus 5 degrees) Fahrenheit. (Caution: When signals are hand fired, gloves should be worn.) Within 30 seconds after removal from the cold chamber, the signals shall be fired by the launcher. Signals that fail to meet the requirements specified in 3.7.12 shall be considered defectives.

4.5.14 High temperature. The signals and the launcher shall be soaked a minimum of 5 hours at plus 160 degrees (plus or minus 5 degrees) Fahrenheit. (Caution: When signals are hand fired, gloves should be worn.) Within 30 seconds after removal from the heat chamber, the signals shall be fired. Signals that fail to function as specified in 3.7.13 shall be considered to be defectives.

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4.5.15 Functioning after immersion in water. Signals that have not been subjected to preconditioning shall be immersed in water to a depth of 11 to 13 inches of water for a minimum 3-hour period at a temperature of plus 70 degrees (plus or minus 10 degrees) Fahrenheit. The signals shall then be subjected to the function test described in 4.5.4.

* 4.5.16 Firing pin force (launcher). A suitable test plug that will simulate the forces required to initiate the primer of the actual signal shall be utilized for this test. With the test plug installed in the launcher and the top of the test plug 1/8 inch from a self locking force gage (catch gage), the fully cocked firing pin shall be released, and the reading shall be recorded. Launchers that fail to meet the requirements of 3.7.15 shall be rejected. The launcher shall be actuated by hand. If the firing pin force of the launcher is not sufficient to simulate the forces required to initiate the primer of the signal or if the firing pin cannot be pulled back to the firing position (release position) with the thumb, the launcher shall be rejected.

* 4.5.17 Ballistic functioning (launcher). The launcher shall fire the signals. Failure of the launcher to fire a minimum of 14 signals successfully without requiring any maintenance of the launcher (no cleaning) shall be cause for rejection. When production launchers are subjected to this test, the signals fired by the production launchers shall be production signals.

* 4.5.18 Vibration (launcher). The launcher shall be subjected to vibration at plus 86 degrees (plus or minus 18 degrees) Fahrenheit, in accordance with method 514, category g, procedure XI, table 514.2-VII, figure 514.2-7 of MIL-STD-810. After the vibration test, the launcher shall launch a minimum of 14 signals without requiring any maintenance of the launcher.

4.5.19 Salt spray (launcher). The launcher shall be subjected to salt spray (fog) exposure for a 50-hour period in accordance with method 509, procedure I of MIL-STD-810. After the salt spray test the launcher shall be cleaned in water. The launcher shall then fire a minimum of 14 signals successfully without requiring any maintenance of the launcher.

* 4.5.20 Extreme high temperature storage. The signals shall be subjected to the high temperature test described in method 501, procedure I of MIL-STD-810 except that the chamber shall be maintained at plus 200 degrees (plus or minus 5 degrees) Fahrenheit for 14 days in lieu of at plus 160 degrees Fahrenheit for 48 hours. The signals shall not burn, explode, or become unsafe to handle during the extreme high temperature storage test. At the conclusion of the extreme high temperature storage test, the signals shall be functioned at ambient temperature and shall not malfunction in a hazardous manner (see 6.5).

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* 4.5.21 Reliability. Successful completion of the preproduction tests specified in 4.3 and table I herein, which include tests related to the functional performance parameters of the signal (see 3.7.3) will demonstrate compliance with the functional reliability requirement specified in 3.5.2.

* 4.5.22 High temperature storage. The signals shall be subjected to a high temperature storage test at plus 160 degrees (plus or minus 4 degrees) Fahrenheit for 28 days as described in test 112 of MIL-STD-331 (excluding the temperature exposure conditioning at minus 65 degrees Fahrenheit). The signals shall then be functioned as described in 4.5.4. Signals that explode, burn, or fail to function as specified in 3.7.20 shall be considered defectives.

4.5.23 Low temperature storage. The signals shall be subjected to a low temperature storage test at minus 65 degrees (plus or minus 4 degrees) Fahrenheit for 28 days as described in test 112 of MIL-STD-331 (excluding the temperature exposure conditioning at plus 160 degrees Fahrenheit). The signals shall then be functioned as specified in 4.5.4. Signals that explode, burn, or fail to function as specified in 3.7.21 shall be considered defectives.

4.5.24 Weight check (signals and launchers). The weight of the signal and the launcher shall be checked to determine compliance with the requirements specified in 3.10.

* 4.5.25 Service life. Fulfillment of the service life requirement shall be verified by analysis and available test data. No tests are required during the preproduction or production phase of procurement to demonstrate this requirement.

* 4.5.26 Compositions. The stability of all compositions shall be verified by testing in accordance with method 501.1, 502.1 and 503.1.1 of MIL-STD-650. Compatibility of all compositions shall be verified by testing in accordance with method 504.1.1 of MIL-STD-650.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A or C. The signal kit and the spare bandoleer assemblies packaged in accordance with level A shall be preserved and packaged in accordance with method IC-3 of MIL-P-116.

5.1.1.1 Intermediate packaging. One-hundred signal kits or 200 spare bandoleer assemblies as specified in 5.1.1 shall be packaged in an interior box that conforms to style RSC, class weather-resistant, grade V3c of PPP-B-636 and that is divided into cells or slots by partitions as specified

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in figure 6. Each of the 100 signal kits shall be placed in the interior box so that only one signal kit will be in each of the 100 cells or slots (see figures 6 and 7). Each of the 200 spare bandoleer assemblies shall be placed in the interior box so that two spare bandoleer assemblies will be in each of the 100 cells or slots (see figure 6). The interior box shall be closed in accordance with the appendix of PPP-B-636.

5.2 Packing. Packing shall be level A or C as specified (see 6.2).

5.2.1 Level A or C. The interior box containing either the 100 signal kits or the 200 spare bandoleer assemblies preserved and packaged as specified shall be packed in an exterior box conforming to overseas type, style A of PPP-B-601 (see figure 6).

5.3 Marking. In addition to any special marking required by the contract or order, the interior box and the exterior box shall be marked in accordance with MIL-STD-129.

5.3.1 Marking drawings. When specified by the procuring activity (see 6.2), marking drawings shall be prepared and submitted, for review, in accordance with MIL-STD-1168. The lot numbering for marking shall be in accordance with MIL-STD-1168.

5.4 Ammunition lot data cards. When specified by the procuring activity (see 6.2), ammunition lot data cards shall be prepared in accordance with MIL-STD-1167 and submitted concurrently with the delivery of the production lots of signal kits and spare bandoleer assemblies.

6. NOTES

6.1 Intended use. The signals covered by this specification are intended to be used by aircrewmen to signal their location as an aid to search and rescue. The signals are intended to penetrate moderately dense jungle foliage and are intended for worldwide application under all anticipated temperature extremes.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification

(b) Specify whether signal kit or spare bandoleer assembly is required and specify color of spare bandoleer assembly (see 1.1, 3.2.2, and 3.11)

(c) Specify preproduction test sample and preproduction and production test data requirements (see 3.1, 4.3, 4.3.1, and 4.4.2)

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(d) Specify applicable levels of preservation and packaging and packing (see 5.1 and 5.2)

(e) Specify requirements for marking drawings and for ammunition lot data cards (see 5.3.1 and 5.4).

6.3 Pyrotechnic. The signals covered by this specification are classified as Common Fireworks, Class C Explosives in accordance with the Code of Federal Regulations 49CFR 171-178.

6.4 General safety precautions. The loading, the assembling, and the handling of the signal kits and of the spare bandoleer assemblies (including their subassemblies) involve hazardous operation; therefore, DOD Manual 4145.26M shall be used as a guide for safety procedures in handling explosive loaded items.

* 6.5 Definitions. Definitions are:

Signal - Assembly of rocket motor and candle (flare)

Candle (flare) - Component of signal that produces the colored display

Rocket motor - Propulsion section of the signal

Display - Red, green, or white pyrotechnic flare

Signals that malfunction in a hazardous manner - Signals that fail to comply with the performance requirements of 3.7.3(a).

6.6 Annotation of changes. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This annotation was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Air Force - 11

Procuring activity:
Air Force - 11

Project No. 1370-F111

Reviewer:
Air Force - 70

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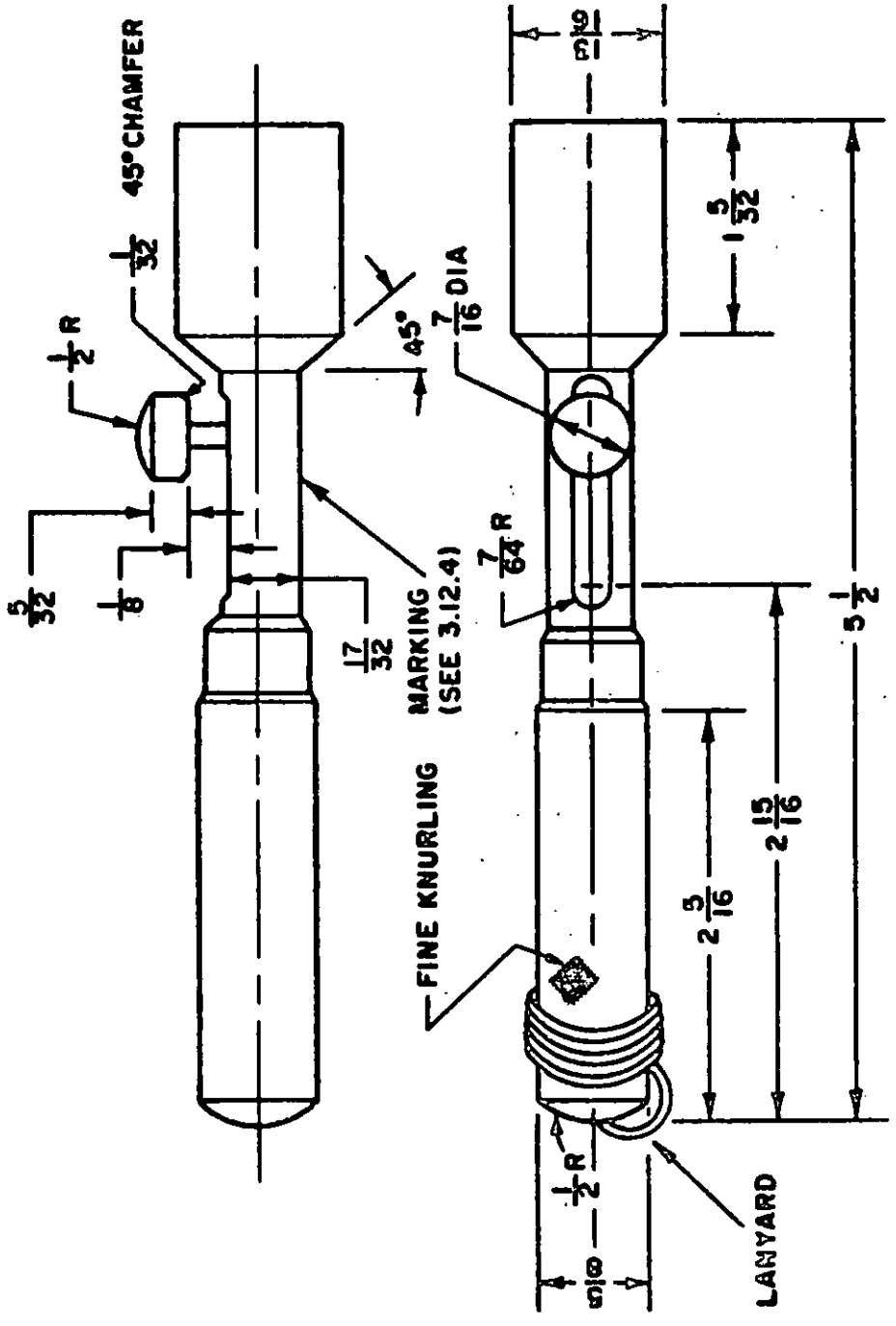
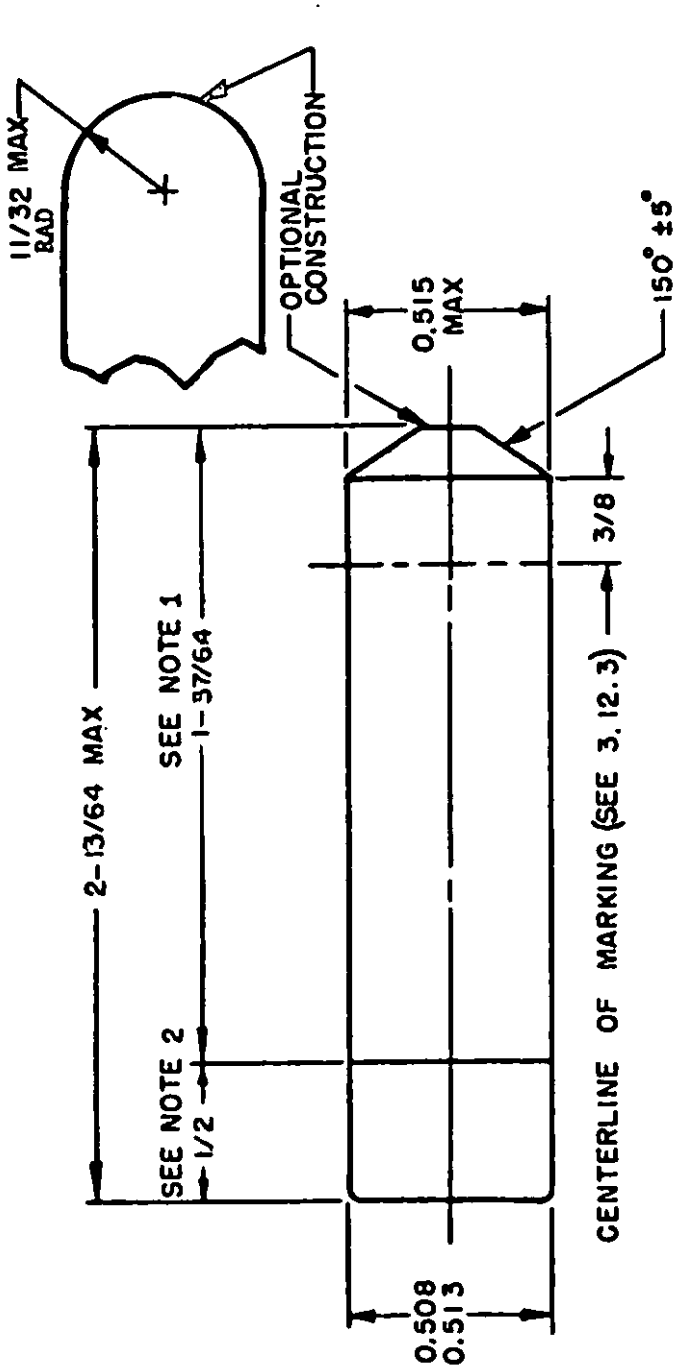


Figure 1. Launcher

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NOTES: 1. PAINTED OR ANODIZED AND DYED SURFACE IN ACCORDANCE WITH 3.6.2 AND 3.11
2. UNPAINTED SURFACE BUT WITH CORROSION-RESISTANT SURFACE
3. UNLESS OTHERWISE SPECIFIED, DIMENSIONS IN INCHES
EXCEPT FOR ANGLES, TOLERANCE ±1/16 INCH

Figure 2. Signal

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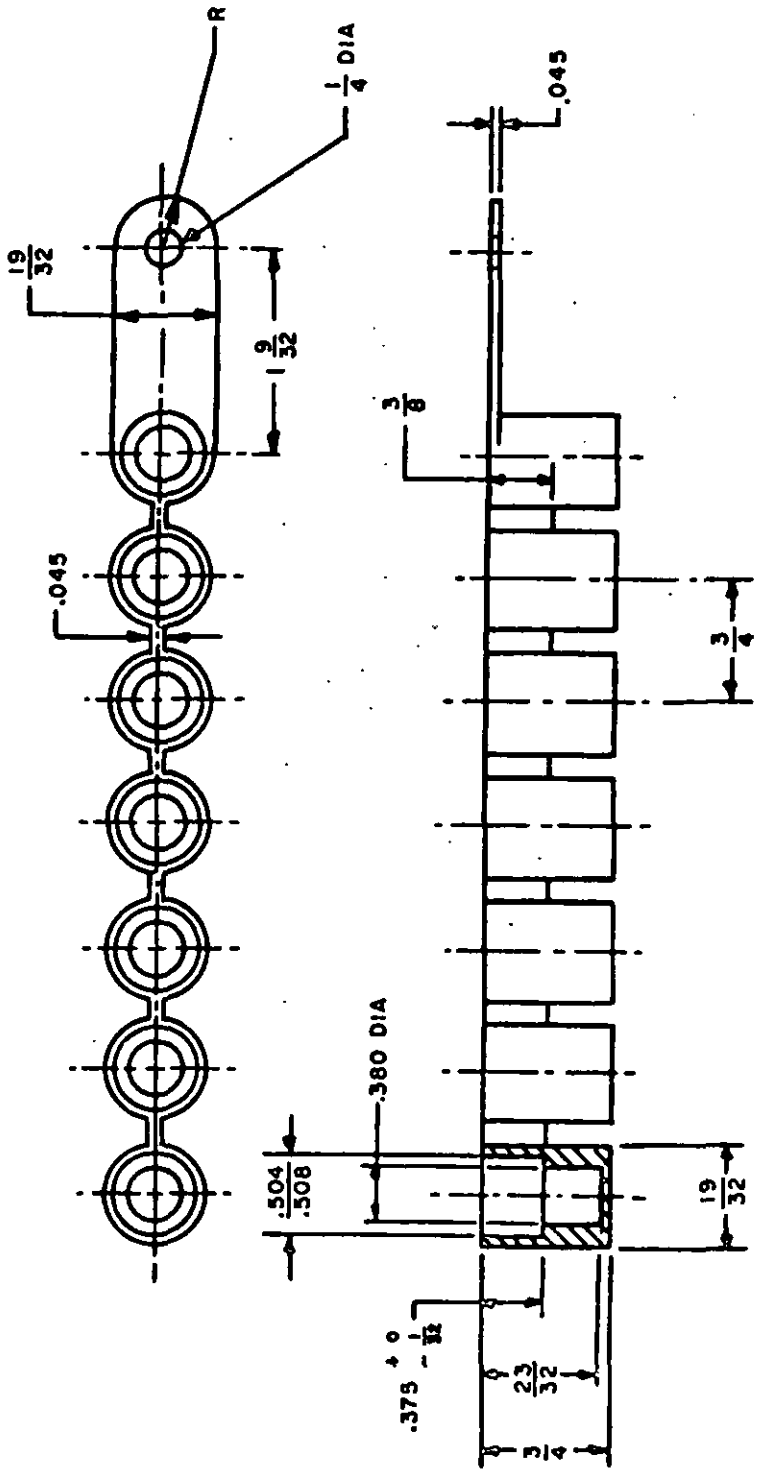


Figure 3. Bandoleer

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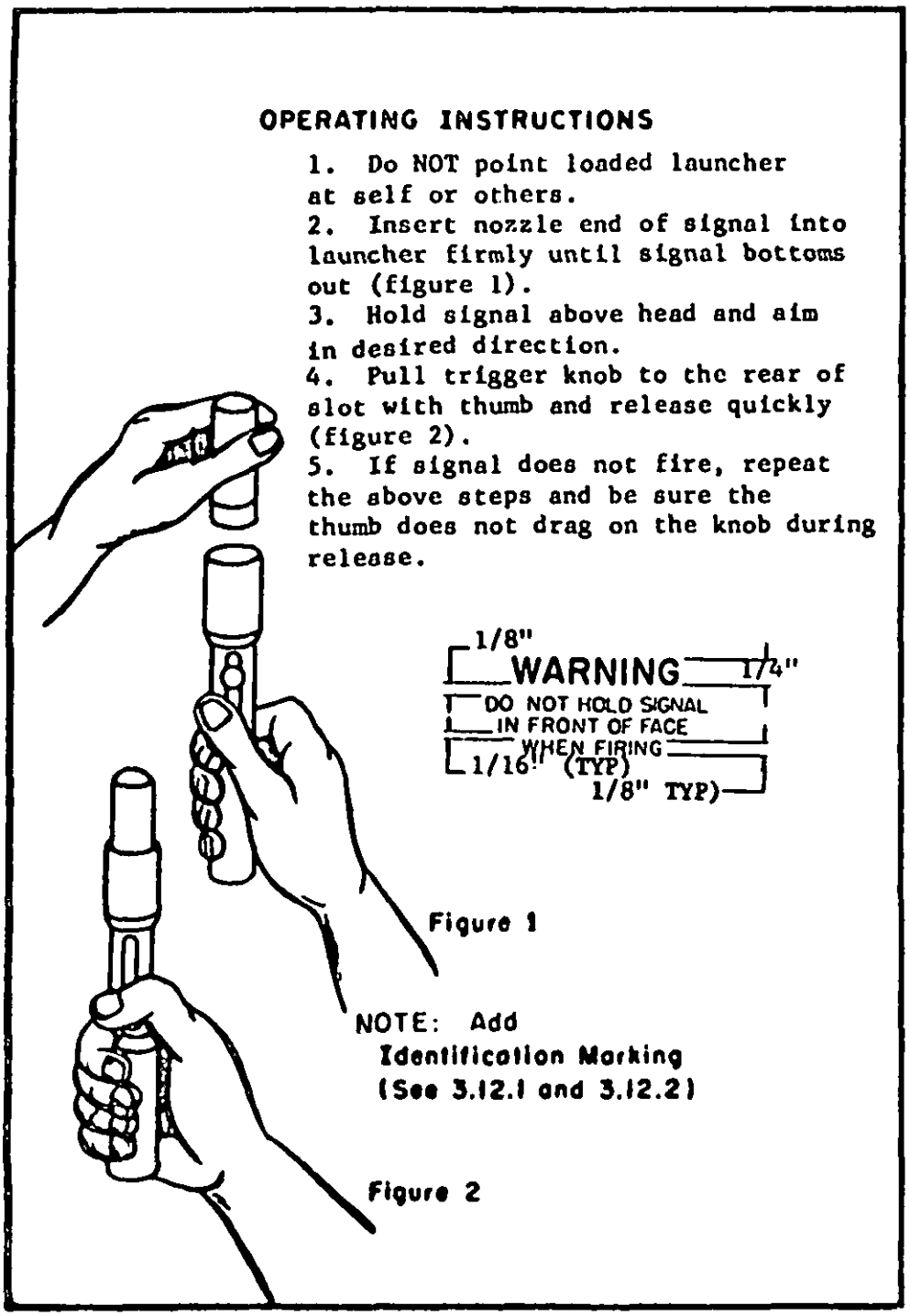
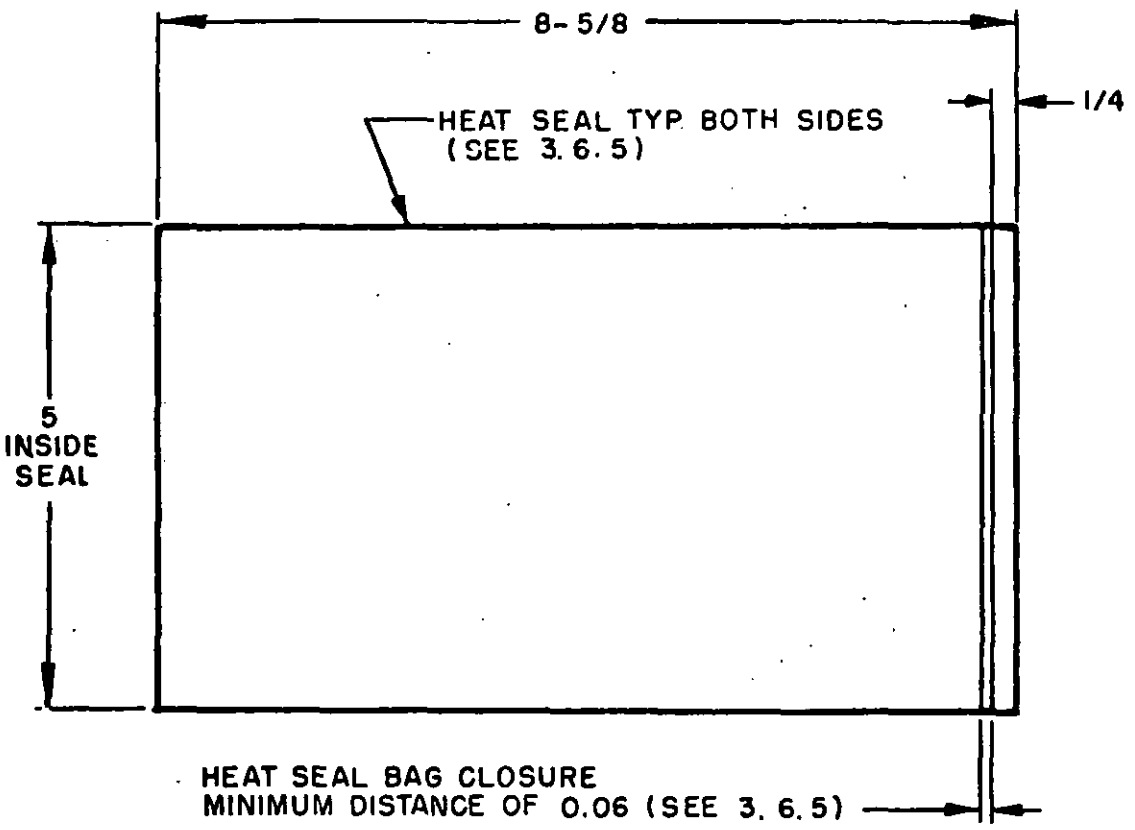
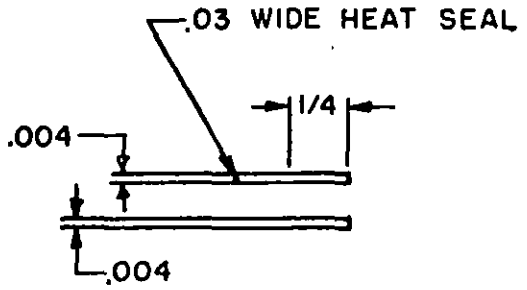


FIGURE 4. Operating Instructions

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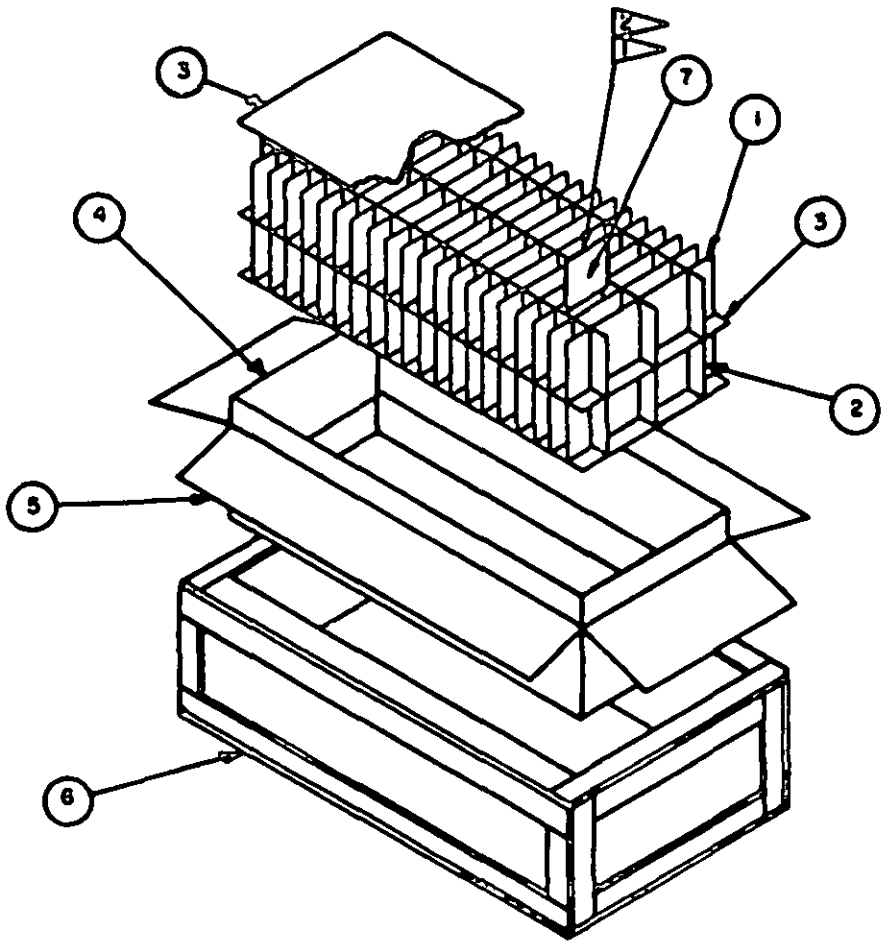




DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED,
TOLERANCES: DIMENSIONS > 2 INCHES ±1/32
DIMENSIONS < 2 INCHES ±.005

Figure 5. Plastic Bag

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MATERIAL FOR PARTITIONS, PADS, AND LINERS: CLASS DOMESTIC, 200-POUND TEST OF PPP-F-320

1 TWO SPARE BANDOLEER ASSEMBLIES IN EACH SLOT IN INTERIOR BOX

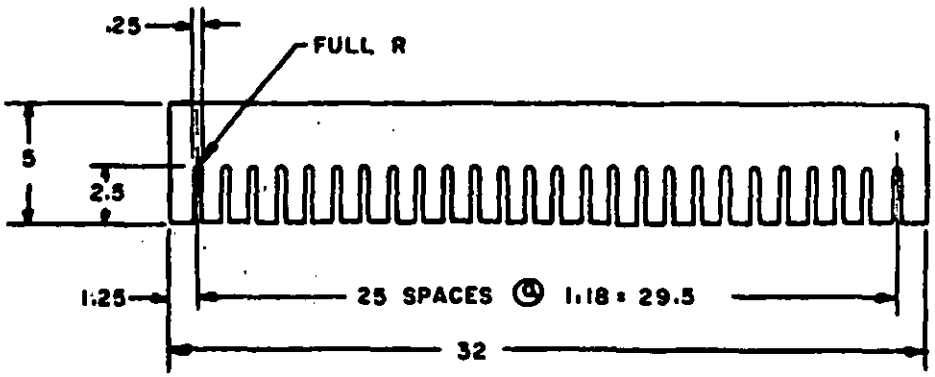
2 ONE SIGNAL KIT IN EACH SLOT IN INTERIOR BOX

COMPONENTS AND APPROXIMATE DIMENSIONS IN INCHES:

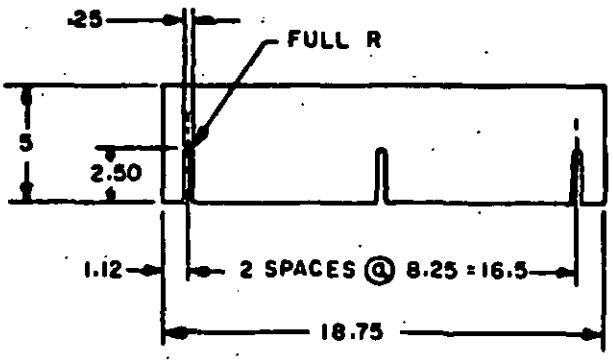
1 PARITITION, 5 BY 18-3/4, 52 REQUIRED (SEE FIGURE 7)
2 PARITITION, 5 BY 32-1/16, 6 REQUIRED (SEE FIGURE 7)
3 PAD, 31-7/8 BY 18-5/8, 3 REQUIRED
4 LINER, 32-1/16 BY 18-3/4 BY 10-3/8, 1 REQUIRED
5 INTERIOR BOX, 32-3/8 BY 19-1/16 BY 10-1/2, 1 REQUIRED
6 EXTERIOR BOX, 32-3/4 BY 19-3/8 BY 11, 1 REQUIRED
7 SIGNAL KIT (100 REQUIRED) or SPARE BANDOLEER ASSEMBLY (200 REQUIRED)

Figure 6. Preparation for Delivery

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SIGNAL KIT



SIGNALS

DIMENSIONS IN INCHES

Figure 7. Partitions

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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